

Kagerou — Feature List

فهرست قابلیت‌ها

Bilingual feature inventory: everything shipped, plus the must-implement roadmap.

دوزبانه: همه قابلیت‌های موجود + نقشه راه مورد نیاز.

Implemented

پیاده‌سازی شده

1. Core GPU Pipeline / GPU

Feature /	Notes /
NVDEC hardware decode	H.264 / H.265, zero-copy to GPU memory
NVENC hardware encode	H.264 / H.265, zero-copy from GPU memory
MP4 + MKV muxing	Minimp4 / FFmpeg, correct durations + audio
FramePool	acquire/release, no per-frame cudaMalloc
CUDA graphs	capture/replay for fixed filter chains
Multi-stream compositor	composite_nv12() — N streams tiled on GPU
Decoder flush	B-frame DPB drain — 600/600 frames, 44+ dB PSNR
Studio-range color	BT.709 limited-range kernels (NVDEC default)
Fully GPU frame path	Frames never leave the device; only tiny result tensors download

2. 24 Non-AI GPU Filters / (-) GPU

Denoise (bilateral + NV12-native) · CLAHE · Gaussian Blur · Sharpen · Brightness/Contrast · Saturation · Gamma · Vignette · Film Grain · Directional Blur · Edge Detect · White Balance · Lens Distortion · Flip · Crop · Pad · Chroma Key · Background Blur · Temporal Denoise · Temporal Stabilization · HDR Tone Map (Reinhard + ACES) · LUT (6 presets + custom .cube) · Super-Res 2x · Frame Blend

تمام فیلترها روی هسته‌های CUDA: بدون درگیری CPU.

All filters run on CUDA kernels; no CPU involvement.

3. 12 AI Filters (ONNX Runtime + TensorRT FP16) /

Filter /	Model /	Description /
Pose Estimation	MediaPipe Pose	33 body landmarks, skeleton
Depth Estimation	Depth Anything V2	near=bright heatmap
Optical Flow	RAFT-Small	Middlebury motion color
AI Denoise	FastDVDNet	temporal, 5-frame window
AI Matting	RVM MobileNetV3	background removal + blur composite
AI Face	YuNet	face boxes, drives autoframe + gaze
AI Hands	BlazePalm + landmarks	21 joints per hand
AI Gaze	MediaPipe iris	iris dots + gaze vectors
AI LowLight	Zero-DCE	dark-room enhance
AI Anime	AnimeGANv3 Hayao	full-frame restyle
AI Detect	YOLOv8n	80 COCO classes, boxes + labels (live/record/transcode)
AutoFrame	(YuNet)	face-tracked crop, Wide/Med/Tight

Gestures (heuristic, no model): CUT · MARK · PAUSE · STOP

Fully GPU: TensorRT EP primary (engines built once at warmup, cached in trt_engines/), CUDA EP fallback per model.

کاملاً روی GPU (FP16: TensorRT EP) اصلی و CUDA EP پشتیبان.

4. Four Tabs /

Tab /	What it does /
LIVE	Virtual camera + screen capture, filters, hotkeys, FPS meter, GPU: HOW IT RUNS dialog
TRANSCODE	File to file with any filter chain; NVDEC -> GPU -> NVENC (H.264/H.265)
RECORD	Live-cut: gesture CUT/MARK, system voice, take.log, cut preview, overlays burned into MP4
GAME	Gesture game control — EXTERNAL (SendInput mouse+keyboard into a detected game window) + TRAINER (in-app shooting gallery with sounds/score). Palm = aim, peace/pinch = fire, flick = jump, fist = reload, open palm = pause, thumb = use. F12 kill switch.

Tool /	What it does /
Virtual camera (DirectShow)	vcam DLL registered system-wide; browser/OBS see it
screen_cap	GPU screen capture for the LIVE feed

Tool /	What it does /
selftest / ai_diag / gametest / kagerou_test	diagnostics, model probes, injector self-check

باقی‌مانده — باید پیاده شود Remaining — must implement

#	Feature /	EN description	
1	NAFNet denoise	Single-frame AI denoise, cheaper than FastDVDNet for weak GPUs	حذف‌نویز تک‌فریمی، سبک‌تر برای GPUهای ضعیف‌تر
2	Studio lighting	Portrait relighting: key / fill / rim light simulation	نورپردازی چهره: کلید لایت / فیل / ریم لایت
3	Audio enhancement	RNNNoise / DPRNN suppression, echo cancellation, voice isolation	بهبود صدا: حذف نویز، حذف اکو، ایزولاسیون صدا
4	RIFE interpolation	Real-time intermediate frames for 2x/4x FPS	اینترپولاسیون فریم برای 2x/4x FPS
5	MiDaS / DPT-Large	Alternative high-accuracy depth (Depth Anything V2 covers basic today)	دقیق‌تر (عمق فعلی با Depth Anything V2 پوشش داده شده) عمق
6	YOLOv8-seg	Person masks + mask-driven auto-frame (bbox detection already done)	اشخاص + auto-frame با ماسک (تشخیص bbox انجام شده) ماسک

حالت گیمر — نسخه اول آماده شد Gamers Mode — v1 shipped

GAME tab drives real PC games with your hands (no driver — plain Win32 SendInput):

تب GAME بازی‌های واقعی PC را با دست کنترل می‌کند (بدون درایور — SendInput استاندارد ویندوز):

GAME tab -- hand tracking (BlazePalm, every 2nd frame while on)

EXTERNAL mode -- foreground game-window detect (GRAB GAME locks exe)

- aim hand centroid = absolute/relative mouse (+smooth, sens, deadzone)
- peace / pinch hold = FIRE (auto while held, debounced)
- flick up = JUMP (150ms tap)
- fist = RELOAD · open-palm tap = PAUSE · thumb = USE
- left-hand zones / face lean = WASD · left pinch = SPRINT
- face drop (sitting) = CROUCH · F12 / open-palm 2s = KILL (releases all)
- bindings editable in UI, saved to game_binds.ini

TRAINER mode -- in-app shooting gallery (camera bg + GPU targets/HUD)

- waves, combo scoring, HP/ammo/timer, pause + game-over menus
- synthesized gunshot/hit/reload (WinMM, no asset files)
- canvas rides g_d_rgb -- preview + vcam + record carry it free

Next (v2 ideas): right-stick aim via second-hand tilt, per-game binding profiles, voice commands for loadout.

قدم‌های بعدی: ایم با شیب دست دوم، پروفایل بایندینگ برای هر بازی، دستورات صوتی.

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